Practical 6 – IP security (IPsec) configuration

Aim:

Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

Theory:

What is IP security (IPsec)?

It is a collection of communication protocols that establish secure connections across a network. It enhances the security of Internet Protocol (IP) by integrating encryption and authentication techniques.

Use of IPsec

It can be used to perform the following functions:

- Protect data transmissions at the router level when using the public Internet.
- It helps in encrypting the application data.
- Quickly verify the authenticity of data if it is sourced from a trusted sender.
- Enhance network data security by establishing encrypted circuits, commonly referred to as IPsec tunnels, which effectively encrypt all transmitted data between two chosen endpoints.
- Safeguard against Replay attack by arranging a unique identifier for each data packet

IPsec modes

IPsec is a network protocol that allows secure communication by offering two modes of operation: tunnel mode and transport mode

- **Tunnel Mode:** Every packet is kept safe when operating in tunnel mode. IPsec encapsulates the data packet within a new packet, applies encryption to it, and adds a new IP header. It is widely utilized for Site-to-site VPN setups.
- **Transport Mode:** The IPsec transport mode encrypts the data packet's payload and keeps the IP header in its original state. The unencrypted packet header helps routers identify the destination address for each data packet.

IPsec protocols

Authentication Header (AH)

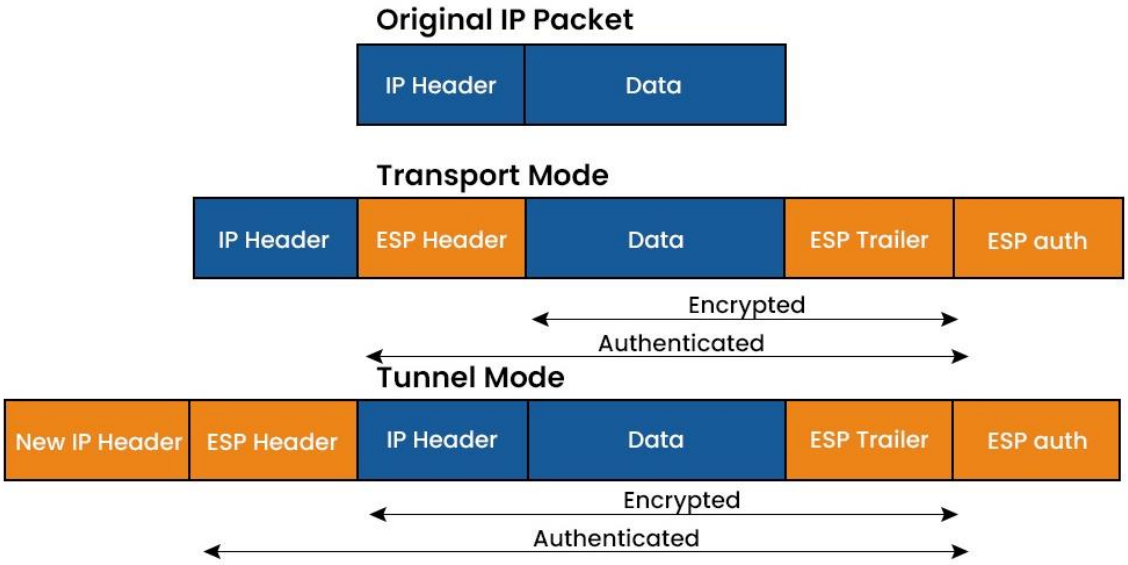
The authentication header (AH) protocol adds a header with sender authentication information to prevent unauthorized parties from compromising a packet's data. It warns the receiver of any modification that may have occurred with the original data packet.

Encapsulating security payload (ESP)

Encapsulating Security Payloads (ESP) provides an additional layer of authentication along with encryption.

In the **transport mode** implementation of IPsec, ESP wraps the data payload while leaving the header intact and accessible.

In **tunnel mode**, ESP envelops the whole data packet and adds an additional header on the outside. The outer header is the only visible component of the system without the need for authentication and the encryption key.



Internet Key Exchange (IKE)

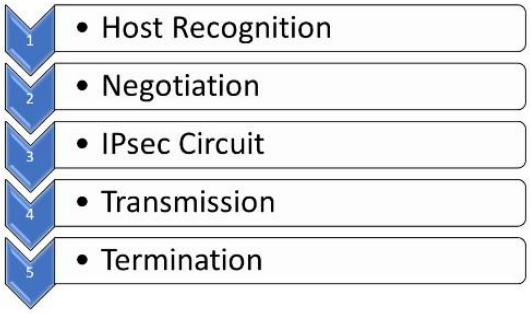
The Internet Key Exchange (IKE) protocol creates a secure connection between two devices operating on the Internet. Both devices establish a security association (SA) by engaging in the process of negotiating encryption keys and algorithms that will be used for the transmission and reception of future data packets.

Internet Security Association and Key Management Protocol (ISAKMP).

ISAKMP is specified as part of the IKE protocol. It is a framework for key establishment, authentication and negotiation of an SA for a secure exchange of packets at the IP layer. The SA includes all attributes of the connection, including the cryptographic algorithm, the IPsec mode, the encryption key and any other parameters related to data transmission over the connection.

Working of IPsec

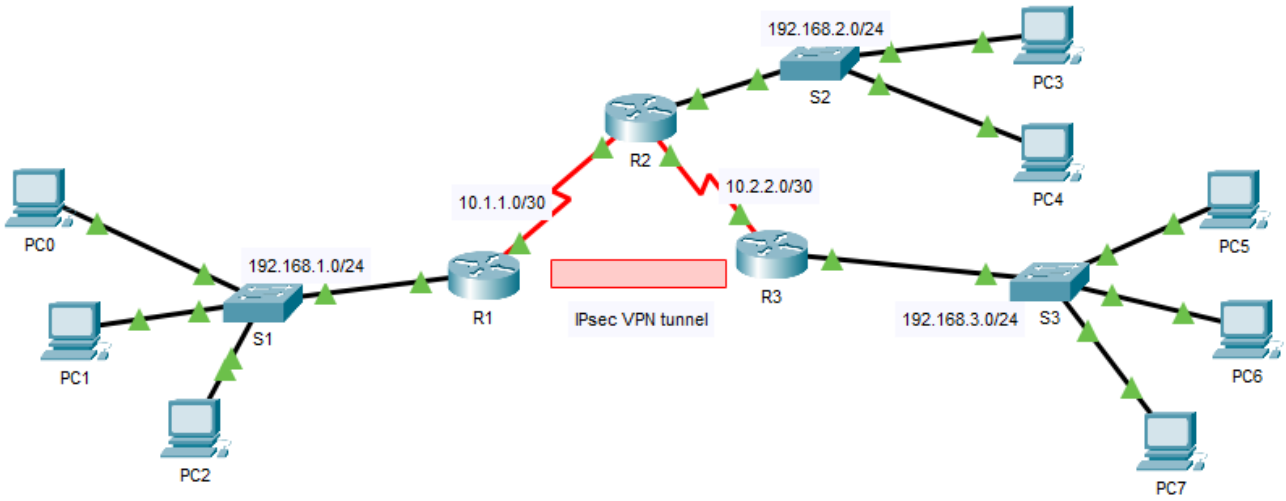
The working of IPsec is classified into five steps. These are shown below with a diagram.



1. **Host recognition.** The IPsec process begins when a host system recognizes that a packet needs protection and should be transmitted using IPsec policies. Such packets are considered "*interesting traffic*" for IPsec purposes, and they trigger the security policies.
2. **Negotiation (IKE Phase 1)** In the second step, the hosts use IPsec to negotiate the set of policies they will use for a secured circuit. They also authenticate themselves to each other and set up a secure channel between them that is used to negotiate the way the IPsec circuit will encrypt or authenticate data sent across it. This negotiation process occurs using either main mode or aggressive mode.
3. **IPsec Circuit (IKE Phase 2)** After successfully establishing a secure connection, as mentioned above, an Internet Protocol Security circuit is then established. The host systems establish a mutual agreement on the encryption and decryption keys to be utilized. Additionally, they exchange cryptographic nonces, which are randomized numbers utilized for authentication purposes.
4. **IPsec transmission.** In the fourth step, the hosts exchange the actual data across the secure tunnel they've established. The IPsec SAs set up earlier are used to encrypt and decrypt the packets.
5. **Termination** The IPsec connection is terminated when either all of the data has been transmitted, or the session duration limit has been reached. The private keys used in the transaction are deleted upon completion of the packet transfer.

Procedure:

1. Open Cisco packet tracer and create a topology as shown in diagram



2. Use following Addressing table to assign IP addresses to various interfaces ad end-points.

Device	Interfaces	IP address	Subnet mask	Default gateway
R1	G0/0	192.168.1.1	255.255.255.0	NA
	S0/0/0 (DCE) (64000)	10.1.1.2	255.255.255.252	NA
R2	G0/0	192.168.2.1	255.255.255.0	NA
	S0/0/0	10.1.1.1	255.255.255.252	NA
	S0/0/1 (DCE) (64000)	10.2.2.1	255.255.255.252	NA
R3	G0/0	192.168.3.1	255.255.255.0	NA
	S0/0/1	10.2.2.2	255.255.255.252	NA
PC0	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.4	255.255.255.0	192.168.1.1
PC3	NIC	192.168.2.2	255.255.255.0	192.168.2.1
PC4	NIC	192.168.2.3	255.255.255.0	192.168.2.1
PC5	NIC	192.168.3.2	255.255.255.0	192.168.3.1
PC6	NIC	192.168.3.3	255.255.255.0	192.168.3.1
PC7	NIC	192.168.3.4	255.255.255.0	192.168.3.1

Objective:

configure R1 and R3 to support a site-to-site IPsec VPN when traffic flows between their respective LANs. The IPsec VPN tunnel is from R1 to R3 via R2. R2 acts as a pass-through and has no knowledge of the VPN.

3. Configure routing protocol – RIP at each router R1, R2 and R3 to establish neighbourship.

4. Test the connectivity – Ping from PC0 to PC7 and vice-versa, it should be successful.

Part 1: Configure security technology package on R1 and R3.

1. Enable the Security Technology package.

a. Assign a hostname R1 to router R1 and issue the show version command in to view the Security Technology package license information.

R1#show version

```
Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      None          None          None
data          None          None          None

Configuration register is 0x2102
```

b. If security feature is not enabled, we need to enable the security package by using following command in configuration mode –

R1#configure terminal
R1(config)#
R1(config)#license boot module c1900 technology-package securityk9

c. Accept the license agreement and reload the router to enable the security license.

R1(config)#exit
R1#
R1#reload

d. Save and confirm the changes in security configuration. Verify that the Security Technology package has been enabled by using the **show version** command.

```
Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
data          disable       None          None

Configuration register is 0x2102
```

The security package is enabled.

e. Similarly enable security package at router R3.

2. Create ACL to identify interesting traffic on R1 and R3.

Configure **Access Control List (ACL) 100** to identify the traffic from the LAN on R1 to the LAN on R3 as interesting. This interesting traffic will trigger the IPsec VPN to be implemented when there is traffic between the R1 to R3 LANs. All other traffic sourced from the LANs will not be encrypted.

R1(config)# access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255

R3(config)# access-list 100 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255

Part 2: Configure IPsec Parameters on R1 and R3 (IKE Phase 1 ISAKMP policy)

1. Configure the IKE Phase 1 ISAKMP policy on R1.

Configure the crypto ISAKMP policy 10 properties on R1 along with the shared crypto key. Refer following table for the specific parameters to configure. Default values do not have to be configured.

Parameters	Parameter options and defaults	R1	R3
Key Distribution Method	Manual or ISAKMP	ISAKMP	ISAKMP
Encryption Algorithm	DES, 3-DES, or AES	AES 256	AES 256
Hash Algorithm	MD5 or SHA-1	SHA-1	SHA-1
Authentication Method	Pre-shared keys or RSA	Pre-shared keys	Pre-shared keys
Key Exchange	Diffie – Hellman group 1, 2, or 5	5	5
IKE SA Lifetime	86400 seconds or less	86400	86400
ISAKMP Key		vpnpass	Vpnpass

A. Enable crypto ISAKMP

R1(config)# crypto isakmp enable

R1(config)# crypto isakmp policy 10

B. Configure **HAGLE** (**H**ash, **A**uthentication, **G**roup, **L**ifetime, **E**ncryption technique) parameters

R1(config-isakmp)#hash sha
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#lifetime 86400
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#exit

C. Set peer and key for the serial interface address at R3

R1(config)#crypto isakmp key vpnpass address 10.2.2.2

2. Configure the same IKE Phase 1 ISAKMP policy on R3.

A. Use **do show history** command in R1 to display a log of the router commands that have been used.

B. Copy the configuration commands upto exit command from router R1 and paste them in CLI of R3 in global configuration mode.

C. Set peer and key for the serial interface address at R1

R3(config)#crypto isakmp key vpnpass address 10.1.1.2

3. Verify configuration of the IKE Phase 1 ISAKMP policy on R1 and R3 in privileged EXEC mode.

R1#show crypto isakmp policy

R3#show crypto isakmp policy

Part 3: Configure IPsec Parameters on R1 and R3 (IKE Phase 2 ISAKMP policy)

1. Configure the IKE Phase 2 IPsec policy on R1 and R3.

A. Create the transform-set named **R1toR3** to use esp-aes for encryption and esp-sha-hmac for hashing at R1 and R3.

R1 (config)#crypto ipsec transform-set **R1toR3** esp-aes esp-sha-hmac

R3 (config)#crypto ipsec transform-set **R1toR3** esp-aes esp-sha-hmac

Use command show crypto ipsec transform-set on R1 and R3 to verify the transform-set creation.

B. Create the crypto map IPSEC-MAP that binds all of the Phase 2 parameters together. Use sequence number 10 and identify it as an ipsec-isakmp map.

```
R1(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
R1(config-crypto-map)#description VPN connection to R3
R1(config-crypto-map)#set peer 10.2.2.2
R1(config-crypto-map)#match address 100
R1(config-crypto-map)#set transform-set R1toR3
R1(config-crypto-map)#exit
```

Similarly create same crypto map IPSEC-MAP at R3 for remote peer 10.1.1.2.

```
R3(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
R1(config-crypto-map)# description VPN connection to R1
R3(config-crypto-map)#set peer 10.1.1.2
R3(config-crypto-map)#match address 100
R3(config-crypto-map)#set transform-set R1toR3
R3(config-crypto-map)#exit
```

2. Configure the crypto map on the outgoing interface.

On R1, bind the crypto map IPSEC-MAP to the outgoing Serial 0/0/0 interface and on R3, to Serial 0/0/1 interface.

R1(config)#int s0/0/0

R1(config-if)#crypto map IPSEC-MAP

*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON

R3(config)#int s0/0/1

R3(config-if)#crypto map IPSEC-MAP

*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON

Verify the binding of cypto map IPSEC-MAP to the respective interfaces by the command – **show crypto map** at R1 and R3.

Part 3: Verify IPsec VPN.

1. Verify the tunnel prior to interesting traffic.

Issue the **show crypto ipsec sa** command on R1. Notice that the number of packets encapsulated, encrypted, decapsulated, and decrypted are all set to 0.

```
#pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
#pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

  local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
```

2. Create interesting traffic.

Ping PC-7 from PC-0.

3. Verify the tunnel after interesting traffic. On R1, re-issue the show crypto ipsec sa command. Notice that the number of packets is more than 0, which indicates that the IPsec VPN tunnel is working.

```
#pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
#pkts decaps: 6, #pkts decrypt: 6, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

  local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
```

4. Create uninteresting traffic.

Ping PC-3 from PC-0.

5: Verify the tunnel.

On R1, re-issue the show crypto ipsec sa command. Notice that the number of packets has not changed, which verifies that uninteresting traffic is not encrypted.

```
#pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
#pkts decaps: 6, #pkts decrypt: 6, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

  local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
```